

DUARTE SANTOS

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RESEARCH INTERESTS

My research focuses on learning-based humanoid robotics, particularly whole-body locomotion, reinforcement learning for control, human-motion retargeting, and sim-to-real transfer. I am especially interested in how humanoid robots can achieve natural and robust movement across different morphologies and environments, and in combining learned physical skills with increasingly general autonomous robot behaviour.

EDUCATION

Instituto Superior Técnico (IST), University of Lisbon *September 2023 – November 2026 (Expected)*
Master of Science | *Electrical and Computer Engineering* *Lisbon, Portugal*

- Current GPA: **18.3 / 20**
- Relevant Coursework: Model Predictive Control, Autonomous Systems, Machine Learning, Computer Vision

KTH Royal Institute of Technology *August 2024 – January 2025*
Exchange Studies | *Systems, Control and Robotics* *Stockholm, Sweden*

- Obtained the maximum grade of **A**
- Relevant Coursework: Reinforcement Learning, Artificial Neural Networks and Architectures, Safe Robot Planning, Introduction to Robotics

Instituto Superior Técnico (IST), University of Lisbon *September 2020 – July 2023*
Bachelor of Science | *Electrical and Computer Engineering* *Lisbon, Portugal*

RESEARCH EXPERIENCE

Humanoid Locomotion Researcher | Instituto Superior Técnico, ISR-Lisboa *March 2025 – Present*
Master's Thesis in Humanoid Robot Locomotion Control and Deployment *Lisbon, Portugal*

- Developed and evaluated a progressive learning-based humanoid locomotion study spanning **legged locomotion, whole-body locomotion, and human-reference-enhanced motion**, investigating locomotion robustness, natural motion, tracking performance, and energetic behaviour.
- Built equivalent training, validation, and deployment pipelines across **MuJoCo/MJX, NVIDIA Isaac, and Webots**, cross-validating learned locomotion behaviours between physics simulators prior to deployment on a physical **Booster T1 humanoid**.
- Deployed the learned controllers on the physical Booster T1, reproducing the characteristic whole-body movements and locomotion behaviours observed in simulation while investigating sim-to-real discrepancies in actuator response, joint dynamics, observations, and low-level control.
- Developed **heightmap-driven stair ascent and descent** environments in which the robot detects terrain transitions, approaches and positions itself before traversal, adapts whole-body posture and swing-foot motion, clears step faces without collision, and deliberately places its feet onto successive treads.
- Extended terrain locomotion across varying stair geometries and high risers through adaptive foothold positioning, swing clearance, gait timing, action authority, motion-reference weighting, and reinforcement-learning curricula, including controlled descent objectives that distinguish deliberate step-down behaviour from uncontrolled gravitational descent.
- Ported the complete locomotion, human-motion retargeting, and stair-negotiation framework to six additional humanoid platforms: **Booster K1, Unitree G1, PAL Robotics Kangaroo, Fourier GR3, AgiBot A3, and Deep Robotics DR02**, enabling a cross-embodiment study of how morphology influences learned humanoid locomotion.

Researcher | FOMO-HODOR Project (ISR-Lisboa & UT Austin) *October 2025 – Present*
Foundation Models for Humanoid Domestic Robots *Lisbon, Portugal / Austin, USA*

- Foundational member of a **six-person joint ISR-Lisboa / UT Austin research team** developing the Booster T1 into a general-purpose humanoid service robot and establishing the software and robotics infrastructure required for autonomous domestic tasks.

- Developed and integrated capabilities spanning **autonomous navigation, manipulation, speech-based human-robot interaction, perception, and task execution**, contributing across multiple components of the physical humanoid robot stack.
- Contributed to the integration of **language-model-based task planning** with perception, navigation, manipulation, and interaction modules for long-horizon domestic service behaviours and RoboCup@Home tasks.
- Participated in **RoboCup@Home 2026 with LisTex United**, contributing to the development and evaluation of a humanoid system performing autonomous domestic-service tasks in international competition.
- Collaborated within the joint Lisbon–Austin research effort involving researchers from ISR-Lisboa and UT Austin, including **Pedro Lima, Peter Stone, and Justin Hart**.

Intelligent Robots and Systems Group, ISR-Lisboa

April 2025 – Present

Member of SocRob@Home Research and Competition Team

Lisbon, Portugal

- Developed and integrated autonomous service-robot capabilities for the **TIAGo mobile manipulator**, contributing across navigation, manipulation, perception, human–robot interaction, and task-level system integration.
- Represented **SocRob@Home at RoboCup@Home**, with responsibility for the development, integration, and execution of the **Restaurant task**, which involved **human-robot interaction, person detection** in an unknown environment, and **pick and place** manipulation.
- Contributed to manipulation and systems-integration development through intensive team hackathons and testing campaigns in preparation for activities and robot demonstrations within **euROBIN — the European Robotics and AI Network**.

Student Research Mentor | Instituto Superior Técnico, ISR-Lisboa

April 2026 – June 2026

Humanoid Robotics Simulation Project

Lisbon, Portugal

- Mentored a student research internship developing an **official-SDK-compatible simulation framework in NVIDIA Isaac Sim for a humanoid robot**, providing technical guidance on simulation architecture, robot integration, and development methodology.

SKILLS

Programming: Python · C++ · C · MATLAB · JAX · PyTorch · LaTeX

Robotics & Simulation: MuJoCo / MJX · NVIDIA Isaac Sim / Isaac Lab · Webots · ROS 2 · Humanoid Robot Deployment · Navigation · Manipulation · Human–Robot Interaction

Robot Learning & Control: Reinforcement Learning · FastTD3 / TD3 · PPO · Whole-Body Locomotion · Human-Motion Retargeting · Reference Motion Tracking · Sim-to-Real · Domain Randomization · PD Control · Model Predictive Control

Robotic Platforms: Booster T1 · Booster K1 · Unitree G1 · PAL Robotics Kangaroo · Fourier GR3 · AgiBot A3 · Deep Robotics DR02 · TIAGo

LANGUAGES

English (C2 – IELTS Academic Band 8.5) · **Portuguese** (Native) · **French** (Elementary)